

Minchan Kim

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EDUCATION

Qingdao Daewon School

Expected 2027

Major Focus: Aviation, Aeronautics & Aerospace Science and Technology

GPA: 3.94 / 4.0 (Unweighted) | SAT (March 2026) | TOEFL iBT (July 2026)

LEADERSHIP & EXPERIENCE

Founder & President — COSMOS (Aero-Engineering Club)

Sep 2025 – Present

Qingdao Daewon School

- Founded and lead a 10-member aerospace engineering club through full theory-design-fabrication-test cycles.
- Directed a solid propellant rocket thrust test stand project across two static-fire tests, deriving peak thrust (190N), total impulse (270.6 N·s), and specific impulse (95.2s) from live sensor data, cross-validated against theoretical simulation.
- Directed Project AEROVIEW, an open-circuit suction wind tunnel: parametric CAD design (FreeCAD, OpenSCAD) comparing airflow over NACA 0012/2412/4412 airfoils; computed blockage ratios (1.4%–14%) to redesign support structures.

Founder & Developer — Tern (flytern.site)

Mar 2026 – Present

Personal Project

- Built a flight-booking web app comparing fares by price, duration, and delay risk using Next.js, the Duffel API, and React Three Fiber.
- Designed a custom glassmorphism design system and integrated live airport weather, layover badges, and EN/KO internationalization.

Vice President — Student Council

2025 – Present

Qingdao Daewon School

- Promoted from the Planning Department, where I organized joint domestic/international student events and coordinated cross-department logistics.

Vice President — Mechanics Club

Mar 2026 – Present

Qingdao Daewon School

- Designed and built a 4-bit ripple carry adder from first principles of digital logic, verified on breadboard with logic-gate ICs.
- Leading development of SmartFlight, a flight-booking web app (Next.js 16, React 19, Firebase, Three.js).

Group Leader — Support Group Program

2024 – 2026

Qingdao Daewon School

- Participated in all four semesterly iterations of the school's peer-mentoring Support Group program through 11th grade; served as group leader for every 11th-grade iteration.

RESEARCH & SELECTED PROJECTS

F-22-Inspired RC Aircraft: Aerodynamic Design & Flight Stability Research

Sep 2025 – Jan 2026

1st Place, Physics Division — School Research Paper Competition

- Designed, built, and flight-tested an F-22-inspired RC aircraft; built a functional wind tunnel for empirical aerodynamic testing and flow visualization.
- Authored a research paper analyzing the correlation between aerodynamic design variables and flight stability, optimizing wingspan and wing profile from empirical lift/drag data.

Automated CIWS-Style Drone Tracking Turret

2025

Personal Project

- Engineered a turret system that autonomously detects and tracks moving drone targets, integrating real-time tracking algorithms with Arduino-based hardware control.

HONORS & AWARDS

- 1st Place, School Science Fair — Qingdao Daewon School (2025 – 2026)

- 1st Place (Physics Division), School Research Paper Competition — Qingdao Daewon School (G11, 2025 – 2026)
- 2nd Place / Top Excellence Award, School-wide Writing Contest — prose essay, “What I’d Want to Tell My 8-Year-Old Self” — Qingdao Daewon School (G10, 2024)

TECHNICAL SKILLS

Languages/Frameworks: TypeScript, JavaScript, Python, Next.js, React, React Three Fiber | **Engineering:** Arduino, FreeCAD, OpenSCAD, Digital Logic, System Architecture | **Other:** REST APIs, Firebase, Tailwind CSS, GSAP

COMMUNITY & SERVICE

- PSYCHOFISH (2024 – Present) — Ran an anonymous KakaoTalk peer-counseling channel; wrote analytical club-magazine pieces on inter-Korean conflict resolution and leadership philosophy.
- Point of View, PR Team (2024) — Designed rules/materials for a school-wide Manito event; led "Random Acts of Kindness" campus clean-up volunteering.
- Cultural Exchange Workshop, Shandong University, Weihai, China (Jul – Aug 2025) — Mandarin language and cross-cultural immersion program with an international cohort.